NITDGP/BTECH/Reg/Odd/2025-26

NATIONAL INSTITUTE OF TECHNOLOGY DURGAPUR

Odd Semester Mid-Term Examination, 2025-26

Course Code: CEC301 — Full Marks: 25

Course Name: Solid Mechanics — Time: 90 Minutes

Instructions: Answer all the questions from Section A and any **two** from **Section B**. Notations have their usual meanings if not mentioned otherwise.

Question No.	Body of the Question	Marks	Mapped CO
	Section A		
1.	A built-in bar ABC is made of two parts. Part AB is made of aluminium. It has length and radius 500 mm and 60 mm respectively. The part BC is made of steel. It has length and radius 400 mm and 30 mm respectively. If the system is subjected to an axial force P = 40 kN at section B. The supports A and C are fully rigid. Calculate the stresses developed in aluminium and steel. Modulus of elasticity of steel and aluminium are 200 GPa and 75 GPa respectively.	6	CO1
2.	A steel shaft circular in cross-section has to withstand a torque of 15×10^3 N.m. If shearing stress is not to exceed 50 MPa and angle of twist has to remain with one degree per 5 m length of the shaft. Find the minimum diameter of the solid shaft. Take shear modulus of elasticity of steel is 80 GPa.	7	CO2
	Section B		
3.	a) Name the major structural components of a building and mention their individual functions.	2	CO2
	Define a beam. Following this definition, draw a schematic diagram of a beam in three dimensions. Hence show the plane of symmetry and neutral surface of the beam and a typical cross section.	4	CO2
4.	Draw line diagram of each of the following beams, identify each support and find out **the support reactions.** Hence, draw **SFD and BMD** in each case. a) Simple supported beam of length L with concentrated downward load P_0 at a distance $L/4$ from the left support. Overhang beam of overall length 4m out of which there is a single overhang of 1m. There is a concentrated moment of 30 kNm at the free end.	3 3	CO1
5.	a) Write down the governing equation of Pure Bending of a beam and explain each term of the equation.	2	CO2
	b) A cantilever beam is of span 4m and circular cross section of diameter 400 mm; it is under a downward *udl* over the entire span. Find out the intensity of the ***udl***, which will produce a maximum bending stress intensity of 20 N/mm². Ignore self-weight of the beam.	4	CO1

Course Outcomes

- CO1: Development of skills for predicting structural behavior of solids under different loads.
- CO2: Knowledge of basics of analysis and design of structural components made of variety of materials.
- CO3: Developing the requisite skill that helps in studying the advanced courses.